

Carlos Hernández Martínez

QA & AI Engineer | Backend Developer

Bridging the gap between AI innovation and software reliability.

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SUMMARY

Computer Engineer specializing in AI and machine learning, with experience building and validating AI agents, automating QA, and developing backend systems with Python and Spring Boot. At Mercadona IT, he works on an AI-agent platform and an integration-testing framework.

PROFESSIONAL EXPERIENCE

QA & AI Engineer Jul. 2025 - Present
[Mercadona IT](#)

- Develops evaluation workflows to validate the correct behavior and outputs of AI agents.
- Builds AI agents for integration-test development.
- Contributes to an integration-testing framework deployed on Okteto.
- Develops backend services for monitoring automated tests and visualizing their reports.

Backend AI Engineer Oct. 2024 - Jun. 2025
[Urobora SL \(AI Startup\)](#)

- Built autonomous agents for task automation and enterprise RAG systems.
- Developed microservices deployed on GCP and Kubernetes, from research to production.

EDUCATION

Bachelor's Degree in Computer Engineering 2021-2025
[Polytechnic University of Valencia \(UPV\)](#)
Specialization in Artificial Intelligence and Machine Learning.

Erasmus in Computer Science 6 months, 2024
[Technical University of Munich \(TUM\)](#)
AI and robotics exchange program with applied research on implicit neural models.

Akademia Bankinter 2023
[Bankinter Innovation Foundation](#)
Program in innovation and technological entrepreneurship.

Diploma in Social Innovation 2019-2020
[CEU San Pablo Valencia](#)
Program in social innovation and corporate responsibility.

SKILLS

Backend & Core Engineering
Python - Java - Spring Boot - Spring Batch

Artificial Intelligence
AI Agents Development - Autonomous Agents Validation - Google ADK - LiteLLM - Claude - Model Context Protocol (MCP) -

PyTorch - TensorFlow

QA & Software Quality

Integration FWK (Spring Boot) - IntegraT - AI QA Automation - User Story to Test Generation - Mutant Testing - SonarQube - Playwright - Postman

DevOps & Cloud

Google Cloud - CI/CD (GitHub Actions, Jenkins, CloudBees) - Spinnaker

PROJECTS

Medical Diagnosis with Deep Learning - Bachelor's Thesis

Built a deep-learning system to classify knee osteoarthritis from medical radiographs using ResNet and EfficientNet with PyTorch and TensorFlow. Achieved 80% binary-classification accuracy through fine-tuning, data augmentation, and cross-domain validation.

Tags: [Python](#) - [PyTorch](#) - [TensorFlow](#) - [Computer Vision](#) - [Healthcare AI](#)

Links: https://github.com/Tempus23/Radiography_TFG

Neural Implicit Models for Robotics - TUM Research

Developed implicit neural models for real-time collision detection in robotic manipulators in collaboration with TUM and KUKA Robotics. Implemented swept-volume models in PyTorch and optimized inference for industrial robotics research.

Tags: [Python](#) - [PyTorch](#) - [Robotics](#) - [Research](#) - [Neural Networks](#)

Links: <https://github.com/Amrsaeed/FastCollisionDetection>

LANGUAGES

Spanish: Native - English: B2/C1 (Cambridge Certificate)